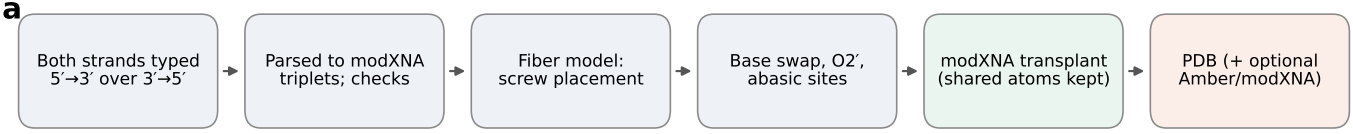
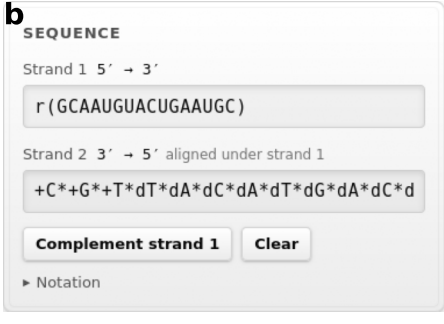
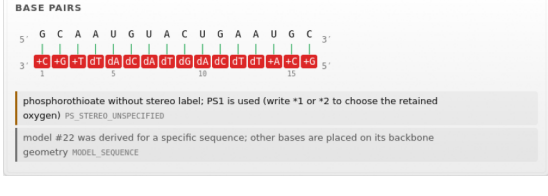
**a**

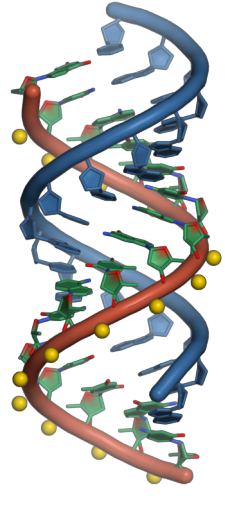



**b**




**c**



RNA (blue) · LNA/DNA gapmer (red)
modXNA residues in sticks, S as spheres

**d** header of the output PDB (REMARK 999 records)

```
dsXNAbuilder 0.2.0-dev
strand 1 (chain A) 5'-r(GCAAUGUACUGAAUGC)-3'
strand 2 (chain B)
    3'-+C*+G*+T*dT*dA*dC*dA*dT*dG*dA*dC*dT*dT*+A*+C*+G-5'
fiber model #22 DNA:RNA hybrid: Hybrid poly (A) : poly d(T)
helical twist 32.7 deg/bp, rise 2.56 A/bp, right-handed
modified nucleotides (modXNA; Love et al. (2024) JCTC 20:9354):
  X01 = 5PO/LNA/RGG 5cap: guanine, LNA (2'-O,4'-C-methylene) ribose,
        5'-OH; 5'-terminal
  X02 = PS1/LNA/RCC int: cytosine, LNA (2'-O,4'-C-methylene) ribose,
        phosphorothioate PS1 (S on OP1 position); internal
  …
modXNA 1.9.3 @ad2e8bd
heavy atoms only (add hydrogens with your modeling or MD program)
ATOM      1  O5'   G A   1       4.535  -7.529 -15.926  1.00  0.00
ATOM      2  C5'   G A   1       5.845  -8.116 -16.028  1.00  0.00
ATOM      3  C4'   G A   1       6.897  -7.028 -16.115  1.00  0.00
…
```